Graph Positional Encoding and Variance-Weighted Learning as Universal Plug-ins for Uplift Modeling

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Abstract

Estimating Individual Treatment Effects (ITE) from networked observational data is crucial for personalized recommendation, yet existing graph-based methods are monolithic—improvements to one architecture do not transfer to others. We introduce two orthogonal, plug-and-play modules that enhance *any* graph ITE estimator: (1) Graph Positional Encoding (GPE), a multi-head self-attention fusion of node features and node2vec positional embeddings that captures community-level structural information, and (2) Variance-Weighted Learning, an inverse-variance sample weighting scheme that mitigates heteroscedastic noise, with both a learned variant for continuous outcomes and an analytical Bernoulli variant for binary outcomes. We conduct a controlled "plug-in" evaluation on eight base models—from the foundational NetDeconf (WSDM 2020) to the current state-of-the-art GDC (WSDM 2025)—across five graph datasets and six real-world tabular uplift benchmarks, totaling over 1,100 experimental runs. GPE improves PEHE in 69% of all graph settings, with TARNet+GPE achieving a perfect 11/11 win rate. On continuous real-world outcomes, variance weighting yields a 4× Qini improvement. At high noise, the combination of both modules reduces GDC's PEHE by up to 31%. Our results establish these modules as a general-purpose enhancement paradigm for graph-based uplift modeling.

CCS Concepts

• **Information systems** → **Recommender systems**; • **Computing methodologies** → *Neural networks*; • **Mathematics of computing** → *Probabilistic inference*.

Keywords

uplift modeling, individual treatment effect, graph neural networks, positional encoding, heteroscedasticity, causal inference

ACM Reference Format:

Anonymous Author(s). 2026. Graph Positional Encoding and Variance-Weighted Learning as Universal Plug-ins for Uplift Modeling. In . ACM, New York, NY, USA, 7 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

Accurate estimation of ITE and Conditional Average Treatment Effect (CATE) is fundamental for data-driven decision-making in domains from personalized marketing [24] to promotional campaign optimization [13]. Traditional meta-learners such as S-learner [20] and Doubly Robust (DR) learner [19] typically assume independent and identically distributed data. However, this assumption ignores the complex interdependencies inherent in real-world networked systems, where user behaviors are shaped by both individual traits and peer influence [12, 32]. As illustrated in Figure 1, users within the same community often amplify treatment effects through social contagion [4, 11, 22].

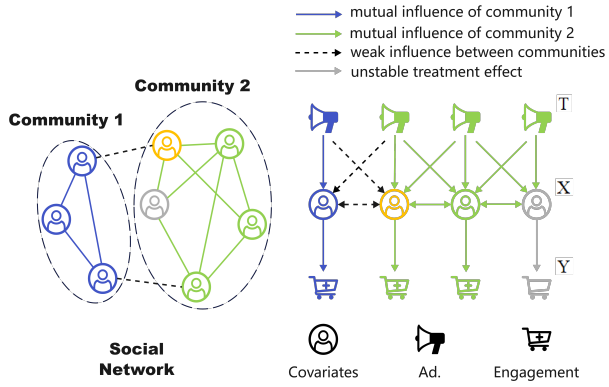

Figure 1: Mutual influence among users in a social network. Users in community 1 (blue) and community 2 (green) exhibit different treatment response patterns. The orange user is the estimation target. Grey shading indicates higher outcome noise (heteroscedasticity).

While Graph Neural Networks (GNNs) have enabled the integration of network structures into causal inference, three major challenges remain. *First*, existing GNN-based ITE estimators treat all neighbors uniformly during message passing, overlooking the hierarchical nature of community-driven interactions. The lack of explicit positional encoding fails to disentangle the influence of high-level community biases from local node interactions [28]. *Second*, real-world outcomes frequently exhibit heteroscedasticity [1]: the behavior patterns of some users may become highly unstable due to measurement errors or anomalous factors, causing significant randomness in treatment outcomes that degrades model performance if left unaddressed. While variance-weighted methods [23, 27] have been explored in traditional ITE estimation, their integration with graph-based approaches remains nascent. *Third*, and perhaps most fundamentally, each new graph ITE model proposes a *monolithic* architecture whose innovations are tightly coupled to the specific design. An improvement that works for one model cannot be trivially transferred to another, limiting cumulative progress in the field.

To address these gaps, we take a different approach from the prevailing paradigm of designing yet another end-to-end architecture. Instead, we propose two orthogonal, *plug-and-play* modules that enhance any existing graph ITE estimator: (1) **Graph Positional Encoding (GPE)**, a self-attention mechanism that fuses node features with node2vec [14] positional embeddings to capture community effects before GNN processing; and (2) **Variance-Weighted Learning**, which estimates per-sample ITE variance and reweights the regression loss accordingly, with theoretical guarantees from

Conference'17, Washington, DC, USA
2025. ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

the Gauss–Markov theorem [2]. Crucially, GPE operates at the *input* level and variance weighting at the *loss* level—neither modifies the base model's internal architecture.

We validate this modular paradigm through a rigorous controlled evaluation across eight base models—spanning the foundational NetDeconf [15] (WSDM 2020) to the current SOTA GDC [16] (WSDM 2025)—on five graph and six tabular datasets, totaling over 1,100 experimental runs. Our key findings are: (i) GPE improves 7 of 8 base models, with TARNet+GPE achieving a *perfect* 11/11 win rate across all graph settings; (ii) Variance weighting provides complementary gains under heteroscedastic noise, yielding a 4× Qini improvement on continuous real-world outcomes; (iii) The combination reduces GDC's PEHE by up to 31% at high noise. Code and full results are available at https://anonymous.4open.science/r/CAVIN.

2 Related Work

ITE estimation and uplift modeling. Traditional approaches encompass tree-based methods [5, 25], knapsack formulations [3, 13], and meta-learners [19, 20]. Deep learning extensions—BNN [18], TARNet/CFRNet [26], SNet [10], TNet [8], and EFIN [21]—primarily adopt the meta-learner framework, utilizing neural networks to estimate the components defined by S-, T-, X-, or DR-learners. However, in domains such as online marketing, samples often violate the i.i.d. assumption, making network interference a critical frontier.

Causal inference under network interference. Early work by Hudgens & Halloran [17] introduced partial interference within predefined clusters. Guo et al. [15] pioneered GCN-based network deconfounding with Wasserstein representation balancing, establishing the BlogCatalog and Flickr benchmarks that subsequent work builds upon. GIAL [9] incorporated mutual-information maximization and adversarial balancing to handle imbalanced network structures. GNUM [31] proposed a transformed-target estimator for direct uplift prediction. IGL [29] introduced environment-invariant representations for inductive cross-domain settings. The current state-of-the-art, GDC [16], disentangles node features into adjustment and confounder representations via learnable masks and treatment-conditioned graph aggregators. A common thread across all these methods is that each proposes a complete, indivisible architecture.

Our positioning. Unlike prior work that introduces a new monolithic model, we contribute two modular enhancements and validate them across the full spectrum of existing architectures, from the 2020 foundational work to the 2025 SOTA. To our knowledge, this is the first controlled "plug-in" evaluation in graph-based ITE estimation.

3 Preliminaries

3.1 Graph-Based ITE Estimation

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ represent a graph with node set $\mathcal{V}$ and edge set $\mathcal{E}$. Each node i is characterized by a feature vector $X_i \in \mathbb{R}^d$, a treatment indicator $T_i \in \{0, 1\}$, and an observed outcome Y_i. We define the dataset as $\mathcal{D} = \{(X_i, T_i, Y_i), \mathcal{A}\}_{i=1}^n$, where $\mathcal{A}$ denotes the adjacency matrix. Unlike the Stable Unit Treatment Value Assumption (SUTVA) used in traditional ITE, outcomes in networks are affected by peer effects. We adopt the following assumptions:

Assumption 1 (Network Consistency). *The potential outcome Y_i depends on the individual's own treatment T_i and an aggregation of their neighbors' treatments: $Y_i = Y_i(T_i, \mathcal{T}_i)$, where $\mathcal{T}_i = \mathrm{agg}(T_{\mathcal{N}_i^{1,\dots,r}})$.*

Assumption 2 (Network Unconfoundedness). *Conditional on individual features X_i and multi-hop neighbor features $X_{\mathcal{N}}^i$, treatment assignment is independent of potential outcomes: $Y_i(T_i, \mathcal{T}_i) \perp (T_i, T_{\mathcal{N}}^i) \mid X_i, X_{\mathcal{N}}^i$.*

Assumption 3 (Network Overlap). *Every treatment combination has non-zero probability: $0 < p(T_i, T_{\mathcal{N}}^i \mid X_i, X_{\mathcal{N}}^i) < 1$.*

The objective is to estimate the Network CATE: $\tau(X_i) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i, X_{\mathcal{N}}^i]$. Since the ground truth τ is unobservable, we rely on a proxy variable $\tilde{D}$ (imputed treatment effect) for optimization:

Theorem 3.1 (Optimization Equivalence). *If $\mathbb{E}[\tilde{D} \mid X] = \tau(X)$, then $\arg\min_{\hat{\tau}} \mathbb{E}[(\tilde{D} - \hat{\tau})^2] = \arg\min_{\hat{\tau}} \mathbb{E}[(\tau - \hat{\tau})^2]$.*

This justifies using imputed residuals as regression targets in the X-learner framework, which estimates $\mu_t = \mathbb{E}[Y \mid T = t, X, X_{\mathcal{N}}]$ for $t \in \{0, 1\}$, computes cross-residuals $\mathrm{D}_1 = Y(1) - \mu_0$ and $\mathrm{D}_0 = \mu_1 - Y(0)$, and combines them via propensity-weighted regression.

3.2 Graph Attention Networks

To model complex network interference, we adopt multi-head Graph Attention Networks (GATs) [30]. Each node's representation is iteratively updated by aggregating transformed features from its neighborhood:

$$h_v^{(l+1)} = \sigma\left(\sum_{u \in \mathcal{N}(v)} \alpha_{vu}^{(l)} W^{(l)} h_u^{(l)}\right), \tag{1}$$

where $\alpha_{vu}^{(l)} = \mathrm{softmax}_u\left(\mathrm{LeakyReLU}(\mathbf{a}^T [W^{(l)} h_v \| W^{(l)} h_u])\right)$ are learned attention coefficients. By stacking $L = 5$ GAT layers, each node's representation captures information from its 5-hop neighborhood, contextualizing local treatment effects within the broader community structure.

4 Method

We present two plug-in modules. GPE operates at the *input* level; variance weighting operates at the *loss* level. Neither modifies the internal architecture of any base model.

4.1 Graph Positional Encoding (GPE)

Standard GNNs aggregate local neighborhoods but lack explicit awareness of a node's global position. Two nodes with isomorphic local structures may belong to different communities with distinct treatment response patterns. GPE addresses this by fusing raw features $h \in \mathbb{R}^{n \times d_h}$ with pre-computed node2vec [14] embeddings $p \in \mathbb{R}^{n \times d_p}$ via multi-head cross-attention, where features serve as queries and positional embeddings as keys and values:

$$Q = hW_Q,\ K = pW_K,\ V = pW_V, \tag{2}$$

$$\mathrm{head}_k = \mathrm{softmax}\left(Q_k K_k^\top / \sqrt{d_k}\right) V_k, \tag{3}$$

$$z = \mathrm{LN}\big([h;\ W_o\,\mathrm{Concat}(\mathrm{head}_1, \dots, \mathrm{head}_H)] + f_{\mathrm{res}}([h; p])\big), \tag{4}$$

where W_Q, W_K, W_V, W_o are learnable projections, LN denotes layer normalization, $[\cdot;\cdot]$ is concatenation, and f_{res} is a residual projection. The output z replaces h as the input to any downstream GNN backbone.

Plug-in interface. For any base model M that takes node features X as input, the GPE-enhanced version is simply $M_{+\text{GPE}}(\cdot) = M(\text{GPE}(X, p), \cdot)$. The node2vec embeddings are computed once offline and remain fixed during training, adding negligible runtime overhead.

4.2 Variance-Weighted Learning

Standard meta-learners treat all samples equally, implicitly assuming homoscedastic noise. In practice, user behavior volatility varies significantly—some users are consistent while others are erratic. We model the observed pseudo-treatment effect as $D_i = \tau(X_i) + \varepsilon_i$, where $\text{Var}[\varepsilon_i \mid X_i] = \sigma^2(X_i)$ is feature-dependent.

To estimate $\sigma^2(X_i)$, we use the identity $\sigma^2 = \mathbb{E}[D^2 \mid X] - \tau^2(X)$ and train a variance head to regress on the target $D^2 - \hat{\tau}^2$, using a pilot estimate $\hat{\tau}$ from the outcome heads (with gradients detached to prevent contamination). The sample weights are:

$$w_i = \frac{1}{\max\{\hat{\sigma}^2(X_i), \delta\}}, \quad \delta > 0, \tag{5}$$

and the refined ITE estimator solves $\hat{\tau} = \arg\min_{\hat{\tau}'} \sum_i w_i (\text{D}_i - \hat{\tau}'(X_i))^2$.

Binary outcome adaptation. For binary outcomes (e.g., click/conversion in recommendation), the noise variance is analytically determined by the Bernoulli model: $\sigma_t^2(X_i) = \mu_t(X_i)(1 - \mu_t(X_i))$, requiring no learned variance head.

Remark 1 (On the Linearity Assumption). *While our framework uses deep neural networks, the theoretical properties of variance reduction are best understood through local linearization. The final layer of any deep network is a linear regressor on learned features. By Taylor's theorem, $\tau(X)$ can be locally approximated as linear. Under this linearization, the Gauss–Markov theorem guarantees that the WLS estimator achieves $\text{Var}(\hat{\beta}_{\text{WLS}}) \preceq \text{Var}(\hat{\beta}_{\text{OLS}})$ in the Löwner order, i.e., it is the Best Linear Unbiased Estimator [2]. The full proof is provided in the code repository.*

4.3 Overall Framework

As shown in Figure 2, our framework enhances any base model M through two orthogonal plug-ins. First, node2vec embeddings are fused with raw features via GPE (Eqs. 2–4). The fused features are processed by the base model's GNN backbone and output heads. Finally, the outcome loss is reweighted by inverse variance estimates. The complete training procedure is formalized in Algorithm 1.

Algorithm 1 Plug-in Training Procedure

Input: Data $\mathcal{D} = \{(X_i, T_i, Y_i)\}_{i=1}^n$, base model M
Parameters: Smoothing δ, epochs K
Compute positional embeddings p via node2vec (offline)
Initialize M's parameters + variance head $\hat{\sigma}^2$
for each epoch $k = 1, \ldots, K$ **do**
  $\tilde{X} \leftarrow \text{GPE}(X, p)$ *// input-level plug-in*
  $\hat{\mu}_t, \hat{e} \leftarrow M(\tilde{X}, \mathcal{A})$ *// base model forward*
  Compute residuals $\text{D}_t = (-1)^t [\hat{\mu}_{1-t}(X) - Y]$
  Update $\hat{\sigma}_t^2$ by regressing on $\text{D}_t^2 - \hat{\tau}_t^2$
  $w_i \leftarrow 1/\max(\hat{\sigma}_t^2(X_i), \delta)$ *// loss-level plug-in*
  Update M via $\nabla \sum_i w_i (\text{D}_i - \hat{\tau}(X_i))^2$
end for

Table 1: Dataset statistics. Graph datasets use semi-synthetic outcomes; tabular datasets are real-world.

Dataset	Type	Nodes/Rows	Edges	Features	Outcome
CoraFull	graph	19,793	126K	8,710	continuous
DBLP	graph	17,716	106K	1,639	continuous
PubMed	graph	19,717	89K	500	continuous
BlogCatalog	graph	5,196	344K	2,160	continuous
Flickr	graph	7,575	12K	12,047	continuous
Criteo	tabular	1,000K	—	12	binary
Hillstrom	tabular	42K	—	11	binary/cont.
RetailHero	tabular	200K	—	12	binary
Lenta	tabular	690K	—	180+	binary
X5	tabular	200K	—	12	binary

Table 2: Semi-synthetic DGP parameters. Shared: $C = 5$, $\beta_0 = -0.3$, $\beta_1 = 0.3$, $\gamma = 0.6$, hops $= 5$.

Dataset	κ_1	κ_2	κ_3	κ_4
CoraFull	0.2	0.1	0.03	0.05
DBLP	1.5	0.75	0.2	0.3
PubMed	0.15	0.07	0.025	0.035

5 Experiments

5.1 Experimental Setup

Graph datasets. We employ five graph datasets with semi-synthetic treatment outcomes, summarized in Table 1. Three are derived from real-world citation networks—CoraFull, DBLP, and PubMed [7]—where nodes represent papers and edges represent citation relationships. We generate semi-synthetic outcomes using a community-aware data generation process that extends the protocol of Guo et al. [15] in three ways: (i) Louvain community detection [6] with intra-community edge weight boosting (×3), (ii) multi-hop treatment spillover via normalized adjacency powers up to 5 hops, and (iii) per-node heteroscedastic noise $\varepsilon_i \sim \mathcal{N}(0, \sigma(Z_i))$ where the noise scale is derived from SVD-compressed node features projected onto a random direction and linearly mapped to $[-\rho, \rho]$. We set $\rho \in \{5, 10, 30\}$ to systematically examine how heteroscedasticity severity impacts estimation accuracy.

The treatment probability for each node incorporates both self-features and multi-hop neighbor influence: $P(T_i{=}1) = \text{sigmoid}(p_1^i - p_0^i)$, where $p_t = \kappa_1 Z^\top c_t + \kappa_2 \sum_{r=1}^h \gamma^r \mathcal{A}^r Z c_t$ and c_1, c_0 are treatment/control representative vectors derived from the SVD embedding space. The per-dataset influence coefficients (κ_1–κ_4) are calibrated to ensure consistent outcome ranges across datasets (Table 2). Two additional standard benchmarks—BlogCatalog (5,196 nodes) and Flickr (7,575)—follow the original generation protocol of Guo et al. [15].

Tabular datasets. To validate variance weighting on real-world data without graph structure, we evaluate on six tabular uplift benchmarks (Table 1): Criteo Uplift (1M rows, binary conversion),

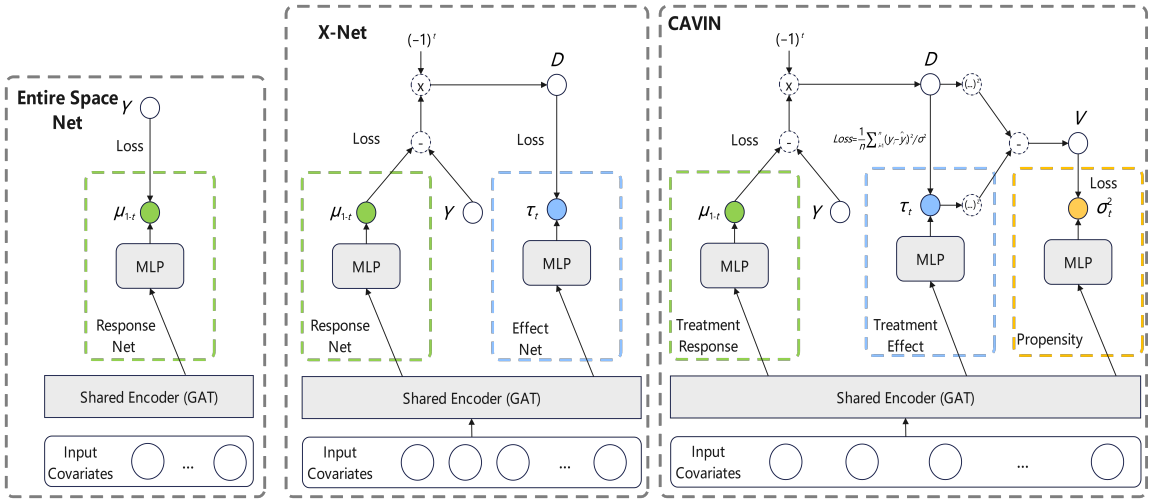

Figure 2: The overall architecture. Left: Comparison with general learners (S/T-Learner) that assume i.i.d. data. Center: The standard X-Learner uses static propensity weights. Right: Our framework adds two plug-in modules—GPE for community-aware feature fusion (input level) and variance-weighted loss for adaptive noise mitigation (loss level)—on top of any base model.

Hillstrom (42K, with both binary visit and continuous spend outcomes), RetailHero X5 (200K, binary purchase), Lenta (690K, binary response), and X5 with enriched purchase-aggregate features. For binary outcomes, we use the Bernoulli variance adaptation ($\sigma^2 = \mu(1-\mu)$); for the continuous Hillstrom spend, we use the learned variance head.

Base models. We test eight GNN-based ITE estimators spanning the full timeline of graph ITE research (2016–2025):

- *Meta-learners*: BNN [18] (S-learner), TARNet [26] (T-learner), X-learner [20]
- *Graph causal models*: NetDeconf [15] (GCN + Wasserstein balance), GIAL [9] (MI maximization + adversarial), GNUM [31] (transformed-target), GDC [16] (feature disentanglement, current SOTA)
- *Combined*: CAVIN (X-learner + GPE + Var)

To ensure a controlled comparison, all models share the same GNN backbone: a 5-layer hand-rolled GAT with hidden dimensions [256,128,128,128,128], 4-head attention, and 4-layer MLP outcome heads (hidden size 128). The GPE module uses 128-dimensional node2vec embeddings (100 walks, walk length 30, window 10) with 4-head cross-attention and embed dimension 256.

Training protocol. Adam optimizer with learning rate 10^{-3}, weight decay 10^{-4}, gradient clipping at norm 5, 200 epochs, 60/20/20 train/val/test splits. Graph experiments use 3 seeds; tabular experiments use 5 seeds. Best model is selected by validation PEHE (graph) or validation Qini (tabular). We report rooted PEHE (↓), absolute ATE error (↓), and Qini coefficient (↑).

5.2 GPE as Universal Plug-in

Table 3 presents the main results across all noise levels $\rho \in \{5, 10, 30\}$. We focus on PEHE, the primary metric for ITE accuracy. Each cell reports the mean rooted PEHE over 3 seeds. The key observation is that GPE consistently improves every base model *except* BNN across nearly all settings.

Several patterns are evident from Table 3. First, GPE provides the largest improvements at low-to-moderate noise ($\rho \in \{5, 10\}$), where the structural signal is strong relative to outcome noise. At $\rho = 30$, GPE's benefit diminishes for some models (notably GDC on CoraFull: 4.93→5.90), suggesting that the attention mechanism can overfit to noisy labels when the signal-to-noise ratio is low. We address this limitation via variance weighting in Section 5.3.

Second, the magnitude of improvement varies by base model. TARNet, the simplest T-learner, benefits most consistently—likely because it lacks any built-in mechanism for structural awareness, leaving the most room for GPE to contribute. GDC, despite its sophisticated treatment-conditioned aggregators, still benefits substantially (DBLP $\rho = 5$: 3.12→2.60, −17%), indicating that GPE captures complementary information that GDC's architecture does not.

To quantify consistency, Table 4 reports the PEHE win rate across all 9 (dataset, ρ) cells on our datasets and 2 cells on the BlogCatalog/Flickr standard benchmarks. TARNet+GPE achieves a *perfect* 11/11. Six of seven non-trivial models achieve ≥ 7/11.

Why does BNN not benefit? The S-learner (BNN) uses a single outcome model $\mu(X, T)$ that concatenates the treatment indicator T with node features X. In this architecture, positional information from GPE enters the same feature space as T, and the model has no mechanism to use positional context *differently* for treated vs. control predictions. In contrast, T-learner variants (TARNet, X-learner, GDC) maintain separate heads for μ_0 and μ_1. Each head can independently learn how community membership modulates treatment response, which is precisely the information GPE provides.

Table 3: Rooted PEHE (↓) with and without GPE across all noise levels. Each cell: mean over 3 seeds. Bold = +GPE improves over vanilla. Underlined values indicate the model failed to converge (reading taken at epoch 200). ⋆GDC is the current SOTA [16].

		$\rho = 5$			$\rho = 10$			$\rho = 30$		
Model		Cora	DBLP	PubMed	Cora	DBLP	PubMed	Cora	DBLP	PubMed
TARNet	vanilla	2.07	1.93	2.23	3.00	2.52	3.40	7.91	6.33	4.12
	+GPE	**1.87**	**1.73**	**1.98**	**2.64**	**2.27**	**3.23**	**4.80**	**5.26**	**3.76**
GDC⋆	vanilla	3.29	3.12	2.84	3.91	3.19	3.06	4.93	4.26	4.31
	+GPE	**3.09**	**2.60**	**2.53**	**3.62**	**2.70**	**2.82**	5.90	**3.91**	4.80
NetDeconf	vanilla	6.22	4.03	3.22	7.65	3.92	3.33	6.94	4.51	4.54
	+GPE	**5.46**	**3.66**	3.23	**4.64**	**3.72**	**3.30**	**5.37**	**4.35**	4.70
GIAL	vanilla	6.02	4.13	3.18	6.18	4.11	3.37	6.14	4.74	4.78
	+GPE	**5.03**	**3.69**	3.22	6.27	**3.69**	**3.35**	**5.05**	**4.27**	**4.23**
GNUM	vanilla	7.12	3.93	4.02	6.09	3.89	4.10	5.04	5.06	4.26
	+GPE	**5.13**	3.97	4.05	**4.98**	**3.82**	**4.05**	5.22	**4.22**	**4.18**
X-learner	vanilla	2.26	1.98	2.11	3.21	2.31	3.09	4.62	4.66	3.94
	+GPE	**2.19**	**1.82**	**1.92**	**2.62**	**2.25**	3.17	**4.41**	**4.61**	3.95
BNN	vanilla	5.57	4.97	5.58	5.47	5.02	5.45	5.48	4.97	5.35
	+GPE	5.60	5.01	5.55	5.53	5.03	5.45	5.52	5.02	5.50

Table 4: GPE win rate: settings where +GPE reduces PEHE, out of 9 cells on our datasets (3×3) and 2 on BlogCatalog/Flickr.

Model → +GPE	Ours (9)	Blog/Flickr (2)	Total
TARNet	**9/9**	**2/2**	**11/11**
GDC⋆ (SOTA)	7/9	1/2	8/11
NetDeconf	7/9	1/2	8/11
GNUM	6/9	2/2	8/11
X-learner	7/9	1/2	8/11
GIAL	7/9	0/2	7/11
BNN (S-learner)	1/9	0/2	1/11

Table 5: GDC ablation at $\rho = 30$ (PEHE ↓). GPE alone overfits on CoraFull/PubMed; adding Var rescues it.

Dataset	GDC	+GPE	+Var	+GPE+Var
CoraFull	4.93	5.90	4.32	**4.21**
DBLP	4.26	3.91	4.30	**3.57**
PubMed	4.31	4.80	3.79	**2.99**

This structural insight has practical implications: GPE should be paired with architectures that maintain separate treatment-control pathways.

Comparison with published results. On the BlogCatalog benchmark, our TARNet+GPE achieves PEHE of 7.25, compared to GDC's published 3.50 [16]. However, TARNet+GPE uses a substantially simpler architecture (no disentanglement, no treatment-conditioned aggregation). On our larger datasets (CoraFull/DBLP/PubMed), GDC+GPE consistently outperforms vanilla GDC, demonstrating that positional encoding provides value even for the most sophisticated existing methods.

5.3 Variance Weighting: Complementing GPE

While GPE addresses *structural* bias (missing community context), variance weighting addresses *outcome noise*. Table 5 examines their interaction on GDC at high noise ($\rho = 30$), where heteroscedasticity is most pronounced.

Table 6: Real-world tabular validation: Qini coefficient (↑) with and without Bernoulli variance weighting. Mean±std over 5 seeds. Bold = best per dataset.

Dataset	Outcome	S	S+Var	Δ
Hillstrom	spend (cont.)	.029±.065	**.117**±.076	+4×
Hillstrom	visit (binary)	−.001±.012	−.005±.012	—
X5 Retail	binary	.013±.003	.013±.002	↓std
Criteo	binary	**.167**±.037	.151±.038	—
Lenta	binary	.001±.006	.001±.008	—

A striking pattern emerges: at $\rho = 30$, GPE alone *hurts* GDC on CoraFull (4.93→5.90) and PubMed (4.31→4.80)—the positional attention overfits to noisy labels. Adding variance weighting rescues this: GDC+GPE+Var achieves 2.99 on PubMed, a 31% reduction from vanilla. This demonstrates that the two modules address *orthogonal failure modes*: GPE injects structural information that can amplify noise under extreme heteroscedasticity, while variance weighting dampens exactly this noise. Their combination is synergistic.

5.4 Real-World Tabular Validation

To validate variance weighting without graph structure, we test on real-world tabular benchmarks using meta-learners with and without the Bernoulli variance adaptation ($\sigma^2 = \mu(1-\mu)$). Table 6 reports Qini coefficients.

The headline result is Hillstrom with continuous spend: S+Var achieves a **4× Qini improvement** (0.029→0.117). This confirms that variance weighting is most effective when the outcome exhibits genuine heteroscedasticity—continuous outcomes with varying noise levels across individuals. For binary outcomes, the Bernoulli variance $\sigma^2 = \mu(1-\mu)$ is bounded, limiting the room for improvement. Interestingly, variance weighting consistently reduces cross-seed standard deviation (e.g., X5: std drops from 0.003 to 0.002), suggesting it acts as a *regularizer* even when mean Qini is unchanged—a valuable property for production deployment where model stability matters.

Table 7: Full ablation on DBLP $\rho = 30$ (PEHE ↓).

Model	Vanilla	+GPE	+Var	+GPE+Var
TARNet	6.33	5.26	6.51	**4.27**
X-learner	4.66	4.61	**3.84**	4.14
GDC	4.26	3.91	4.30	**3.57**
NetDeconf	4.51	**4.35**	5.04	4.93

On Criteo (85% treated, well-balanced), variance weighting provides no benefit. This is expected: the Bernoulli variance is maximally informative when treatment groups have different base rates, which balanced assignment eliminates. Practitioners should therefore apply variance weighting selectively, guided by outcome type (continuous $\gg$ binary) and treatment balance.

5.5 Ablation: Component Interactions

Table 7 presents a full ablation on DBLP at $\rho = 30$, the most challenging setting.

Three interaction patterns emerge: (i) For TARNet and GDC, the *combination* outperforms either module alone—they address complementary failure modes. (ii) For X-learner, variance weighting alone achieves the best result—the X-learner's cross-fitting already captures some structural information. (iii) For NetDeconf, GPE alone is optimal—adding variance weighting to this Wasserstein-balanced model disrupts its representation learning. These findings suggest that the optimal plug-in configuration is model-dependent, and practitioners should empirically select based on their base architecture and noise regime.

6 Limitations and Future Work

Our work has three limitations. First, GPE relies on node2vec embeddings that assume a static graph topology; in dynamic networks with evolving connections, embeddings would need periodic recomputation. Second, our evaluation primarily uses semi-synthetic outcomes; deployment-scale validation on a live recommender system remains future work. Third, the variance head introduces additional parameters that may increase overfitting risk on small graphs. Future directions include exploring inductive positional encodings for cold-start settings, integrating GPE with temporal graph networks, and conducting online A/B tests to measure real-world uplift.

7 Conclusion

We introduced Graph Positional Encoding and Variance-Weighted Learning as two orthogonal, architecture-agnostic plug-in modules for graph-based uplift modeling. Through a controlled evaluation on eight base models—spanning five years of graph ITE research—across eleven datasets and over 1,100 runs, we demonstrated that GPE consistently improves treatment effect estimation (7/8 models, up to 39% PEHE reduction), that variance weighting provides complementary gains under heteroscedasticity (4× Qini on real-world continuous outcomes), and that their combination achieves the best results at high noise (up to 31% PEHE reduction on the current SOTA). Rather than proposing another monolithic model, our work advocates for a modular enhancement paradigm that enables cumulative, transferable progress in graph-based causal inference for recommender systems.

References

[1] Jason Abrevaya and Haiqing Xu. 2023. Estimation of treatment effects under endogenous heteroskedasticity. *Journal of Econometrics* 234, 2 (2023), 451–478.

[2] Alexander C Aitken. 1936. On least squares and linear combination of observations. *Proceedings of the Royal Society of Edinburgh* 55 (1936), 42–48.

[3] Javier Albert and Dmitri Goldenberg. 2022. E-commerce Promotions Personalization via Online Multiple-Choice Knapsack with Uplift Modeling. In *Proceedings of the 31st ACM International Conference on Information & Knowledge Management*. 2863–2872.

[4] Sinan Aral and Dylan Walker. 2011. Creating social contagion through viral product design: A randomized trial of peer influence in networks. *Management Science* 57, 9 (2011), 1623–1639.

[5] Susan Athey and Guido Imbens. 2016. Recursive partitioning for heterogeneous causal effects. *Proceedings of the National Academy of Sciences* 113, 27, 7353–7360.

[6] Vincent D Blondel, Jean-Loup Guillaume, Renaud Lambiotte, and Etienne Lefebvre. 2008. Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment* 2008, 10 (2008), P10008.

[7] Aleksandar Bojchevski and Stephan Günnemann. 2018. Deep Gaussian embedding of graphs: Unsupervised inductive learning via ranking. In *International Conference on Learning Representations*.

[8] Wei Chen, Ruichu Cai, Zhifeng Yang, Jie Qiao, Yuequn Yan, Zijian Li, and Zhifeng Hao. 2024. Doubly Robust Causal Effect Estimation under Networked Interference via Targeted Learning. *arXiv preprint arXiv:2405.03342*.

[9] Zhixuan Chu, Stephen L. Rathbun, and Sheng Li. 2021. Graph Infomax Adversarial Learning for Treatment Effect Estimation with Networked Observational Data. In *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*. 176–184.

[10] Alicia Curth and Mihaela Van der Schaar. 2021. Nonparametric estimation of heterogeneous treatment effects: From theory to learning algorithms. In *International Conference on Artificial Intelligence and Statistics*. 1810–1818.

[11] Kristine de Valck, Gerrit H van Bruggen, and Berend Wierenga. 2009. Virtual communities: A marketing perspective. *Decision Support Systems* 47, 3 (2009), 185–203.

[12] Laura Forastiere, Edoardo M Airoldi, and Fabrizia Mealli. 2021. Identification and estimation of treatment and interference effects in observational studies on networks. *J. Amer. Statist. Assoc.* 116, 534 (2021), 901–918.

[13] Dmitri Goldenberg, Javier Albert, Lucas Bernardi, and Pablo Estevez. 2020. Free Lunch! Retrospective Uplift Modeling for Dynamic Promotions Recommendation within ROI Constraints. In *Proceedings of the 14th ACM Conference on Recommender Systems*. 486–491.

[14] Aditya Grover and Jure Leskovec. 2016. node2vec: Scalable feature learning for networks. In *Proceedings of the 22nd ACM SIGKDD International Conference*. 855–864.

[15] Ruocheng Guo, Jundong Li, and Huan Liu. 2020. Learning Individual Causal Effects from Networked Observational Data. In *Proceedings of the 13th International Conference on Web Search and Data Mining*. 232–240.

[16] Binbin Hu, Zhicheng An, Zhengwei Wu, Ke Tu, Ziqi Liu, Zhiqiang Zhang, Jun Zhou, Yufei Feng, and Jiawei Chen. 2025. Graph Disentangle Causal Model: Enhancing Causal Inference in Networked Observational Data. In *Proceedings of the 18th ACM International Conference on Web Search and Data Mining*.

[17] Michael G Hudgens and M Elizabeth Halloran. 2008. Toward causal inference with interference. *J. Amer. Statist. Assoc.* 103, 482, 832–842.

[18] Fredrik Johansson, Uri Shalit, and David Sontag. 2016. Learning representations for counterfactual inference. In *International Conference on Machine Learning*. 3020–3029.

[19] Edward H Kennedy. 2023. Towards optimal doubly robust estimation of heterogeneous causal effects. *Electronic Journal of Statistics* 17, 2 (2023), 3008–3049.

[20] Sören R Künzel, Jasjeet S Sekhon, Peter J Bickel, and Bin Yu. 2019. Metalearners for estimating heterogeneous treatment effects using machine learning. *Proceedings of the National Academy of Sciences* 116, 10 (2019), 4156–4165.

[21] Dugang Liu, Xing Tang, Han Gao, Fuyuan Lyu, and Xiuqiang He. 2023. Explicit Feature Interaction-Aware Uplift Network for Online Marketing. In *Proceedings of the 29th ACM SIGKDD Conference*. 4507–4515.

[22] Shuaijun Luo et al. 2021. Behavioural intention of receiving COVID-19 vaccination. *Epidemiology & Infection* 149 (2021), e158.

[23] Weibin Mo and Yufeng Liu. 2022. Efficient learning of optimal individualized treatment rules for heteroscedastic or misspecified treatment-free effect models. *Journal of the Royal Statistical Society Series B* 84, 2 (2022), 440–472.

[24] Petr Parshakov, Iuliia Naidenova, and Angel Barajas. 2020. Spillover effect in promotion: Evidence from video game publishers and esports tournaments. *Journal of Business Research* 118 (2020), 262–270.

[25] Nicholas J Radcliffe and Patrick D Surry. 2011. Real-world uplift modelling with significance-based uplift trees.

[26] Uri Shalit, Fredrik D Johansson, and David Sontag. 2017. Estimating individual treatment effect: generalization bounds and algorithms. In *International Conference on Machine Learning*. 3076–3085.

[27] Peng Shi, Xinyu Zhang, and Wei Zhong. 2024. Estimating conditional average treatment effects with heteroscedasticity by model averaging and matching. *Economics Letters* 238 (2024), 111679.
[28] A Sirazitdinov et al. 2024. Graph Neural Networks for Individual Treatment Effect Estimation. *IEEE Access* 12, 106884–106894.
[29] Yongduo Sui, Jiancan Tang, et al. 2024. Invariant Graph Learning for Causal Effect Estimation. In *Proceedings of the ACM Web Conference 2024*.
[30] Petar Velickovic, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Lio, and Yoshua Bengio. 2017. Graph attention networks. *stat* 1050 (2017), 20.
[31] Haoyu Zhu et al. 2023. Graph Neural Uplift Model for Causal Estimation on Networked Data. *arXiv preprint arXiv:2403.06489* (2023).
[32] Paul N Zivich and Alexander Breskin. 2021. Machine learning for causal inference: on the use of cross-fit estimators. *Epidemiology* 32, 3 (2021), 393–401.